

1. Matrix arithmetic / Operations

- Add, Subtract, Multiply, Inverse, Transpose, Hermitian
- $AA^T = A^T A = I$ $\hookrightarrow (AB)^T = B^T A^T, (A^T)^{-1} = (A^{-1})^T$
- Inner prod $\underline{x}^T \underline{y}, \langle \underline{x}, \underline{y} \rangle \rightarrow \text{scalar}$
- Outer prod $\underline{x} \underline{y}^T \rightarrow \text{matrix (rank 1, inverse = 0)}$

2. Solving $\underset{m \times n}{A} \underset{n \times 1}{\underline{x}} = \underset{m \times 1}{\underline{b}}$ (linear eq.)

a) Gauss elim $\rightarrow$ REF

b) Gauss-Jordan $\rightarrow$ RREF (Add back subst.)

#Pivot rows = #Pivot columns = rank(A) = $r \leq \min(m, n)$

#dependent var

#nonpivot col = #independent var

3. Solution set for $A\underline{x} = \underline{b}$

a) Possible sol set:

- ① Unique sol
- ② No sol $\rightarrow \underline{b}$ not lying in Col(A) $\rightarrow$ least square
- ③ Infinitely many sol

b) Form of solution

$$\underline{\hat{x}} = \underline{\hat{x}}_{\text{par}} + \underline{\hat{x}}_{\text{homo}}$$

$$A \underline{\hat{x}}_{\text{par}} = \underline{b}$$

$$A \underline{\hat{x}}_{\text{homo}} = \underline{0}$$

$$\xrightarrow{\text{Span}} \{ \underline{\hat{x}}_{h,s,1}, \underline{\hat{x}}_{h,s,2}, \dots, \underline{\hat{x}}_{h,s,n-r} \}$$

$\hookrightarrow$ Special homogeneous sol $\rightarrow$

4. Special matrices

a) Identity matrix: I_n

b) Diagonal matrix: $D = \begin{bmatrix} a_{11} & & 0 \\ & \ddots & \\ 0 & & a_{nn} \end{bmatrix}$

c) Lower-triangular matrix, $L = \begin{bmatrix} \times & & 0 \\ & \times & \\ 0 & & \times \end{bmatrix}$

d) Upper-triangular matrix, $U = \begin{bmatrix} \times & & \\ & \times & \\ 0 & & \times \end{bmatrix}$

e) Idempotent matrix, $A^2 = A$

ex. projection matrix: $P = A(A^T A)^{-1} A^T$ (Use 0, 1 as eigenvalues)

f) Real-symmetric matrix, $A_{n \times n}, A = A^T$

g) Permutation matrix (permuted rows/cols of I_n) (also orthogonal: $P^T P = I_n, P^T = P^{-1}$)

h) Orthogonal matrix $Q: [\underline{q}_1, \dots, \underline{q}_n]$

$$\hookrightarrow Q^T Q = Q Q^T = I_n \rightarrow Q^{-1} = Q^T$$

5. Elementary row operations $\rightarrow \det(E_1 A) = -\det(A)$
- | | | | |
|----------|-------------------|---------------------|---|
| Type I | R_{ij} | $I \rightarrow E_1$ | $\left. \begin{array}{l} \det(E_1) = -1, \det(E_2) = \alpha \rightarrow \det(E_2 A) = \alpha \det(A) \\ \text{Invertible (Their inverse are also elementary matrices)} \\ \det(E_3) = 1 \rightarrow \det(E_3 A) = \det(A) \end{array} \right\}$ |
| Type II | αR_i | $I \rightarrow E_2$ | |
| Type III | $\beta R_i + R_j$ | $I \rightarrow E_3$ | |

6. Vector spaces / Subspace

- a) Generalized VS
- ① Additive closure
 - ② Scalar Multiplicative closure
 - ③ 8 Axioms

Ex: $\mathbb{R}^n, P_n, C[a, b]$ (continuous func)

b) Subspaces: ① $0 \in W$

② Addition / Scalar multiplication closure on W

(Obtained by $\text{Span } U = \{u_1, \dots, u_m\}$)

c) Linearly independence

$$c_1 u_1 + c_2 u_2 + \dots + c_n u_n = 0$$

Verify if there is a trivial solution or not $\rightarrow$ Only trivial sol = linearly independent
Nontrivial sol = linearly dependent

d) Fundamental subspaces ($\mathbb{R}^n, \mathbb{R}^m$)

• Null space $\text{Null}(A) : N(A)$

$$\text{Null}(A^T) : N(A^T)$$

• Column space, $\text{Col}(A) \in \mathbb{R}^m = \text{Row}(A^T)$

• Row space, $\text{Row}(A) \in \mathbb{R}^n = \text{Col}(A^T)$

• Orthogonal complement $\rightarrow \text{Null}(A) = (\text{Col}(A^T))^\perp = (\text{Row}(A))^\perp$
 $\text{Null}(A^T) = (\text{Col}(A))^\perp = (\text{Row}(A^T))^\perp$

Important observations

$$1) \dim(\text{Col}(A)) + \dim(\text{Null}(A)) = n \quad : \text{Rank-nullity } (A)$$

$$\uparrow \text{rank}(A) \quad \uparrow \text{nullity}(A)$$

$$2) \dim(\text{Row}(A)) = \dim(\text{Col}(A)) = \text{Rank}(A) \quad (\# \text{pivot row/col})$$

$$\leq \min(m, n) \quad [\text{equality} = \text{full-rank}]$$

3) If S is a subspace of $\mathbb{R}^n$, $\dim(S) + \dim(S^\perp) = n$

7. Basis / Dimension

a) Basis of VS V

• linearly independent vectors $\{u_1, \dots, u_n\}$

Newly added $\rightarrow$ linear combi of the rest

• $\text{Span } \{u_1, \dots, u_n\} = V$

Remove one $\rightarrow$ cannot span V

b) $\{u_1, \dots, u_n\} \xrightarrow{\text{Gram-Schmidt}} \{v_1, \dots, v_m\}$

Linearly independent

Orthonormal

c) Dimension of V : $\dim(V) = \# \text{vectors in its basis}$ ($\dim(\{0\}) = 0$, $\{0\}$ doesn't form basis)

8. Determinant ($A_{n \times n}$)

a) $\det(A), |A| \rightarrow$ Determine if A invertible or not

b) $\det(A - \lambda I) = 0 \Rightarrow \lambda_1, \dots, \lambda_n$ eigenvalues

c) Cofactor expansion (row or column)

d) Some useful properties

I. $\det(A) = \det(A^T)$

II. $\det(A) = 0$ if there is zero row/column

III. $\det(AB) = \det(A)\det(B)$

$\det(ABC) = \det(A)\det(B)\det(C)$

IV. $\det(\alpha A) = \alpha^n \det(A)$

V. $\det(A) = \prod_{i=1}^n \lambda_i \leftarrow \text{eigenvalues of } A \quad \therefore \det(A) = 0 \text{ if } \lambda_i = 0 \text{ } \forall i \in \{1, 2, \dots, n\}$

9. Linear transformation

Def $L(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 L(v_1) + \alpha_2 L(v_2)$

Properties: If L is LT, then there must be an $m \times n$ matrix A s.t. $L(x) = Ax$

Change of basis: $[L(u)]_w = A[u]_v$

$\rightarrow$ Similarity of A, B , namely $B = S^{-1}AS$; S : transition matrix

10. Least square: $Ax = b$ has no sol (mainly tall matrix $A_{m \times n}, m \gg n$)

$\arg\min_x \|Ax - b\|^2 \rightarrow x_{ls}$

$A^T A x = A^T b$ normal equation

unique $\hat{x}_{ls}$ for full rank $A \quad \therefore \hat{x}_{ls} = (A^T A)^{-1} A^T b$

Projection $P = A \hat{x}_{ls} = A(A^T A)^{-1} A^T b$ (P idempotent)

Residual: $b - A \hat{x}_{ls} = b - P b = (I - P)b$

11. Inner-product space

a) $\langle x, y \rangle$ x, y elements in GVS

b) Valid I.P. space: $\langle x, x \rangle \geq 0$

$\langle x, y \rangle = \langle y, x \rangle$

$\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$

c) $\mathbb{R}^n, P_n, C[a, b]$

d) Pythagorean's law / Cauchy-Schwarz ineq.

e) $\langle x, y \rangle = 0 \Rightarrow x, y$ orthogonal

$\Rightarrow$ V.S. orthogonal subspace, orthogonal set $\rightarrow$ orthonormal basis for V.S

12. Quadratic equation/form

a) $f(x) = x^T A x + B x + c = 0$ (Quadratic eq.)

Quadratic form

positive definite matrix

positive semi-definite matrix

- b) properties of PD matrix (real symmetric)
- I. PD $\rightarrow$ all eigenvalues positive
 - II. PD $\rightarrow$ leading principal submatrices are also PD.
 - III. PD $\rightarrow A = LU$ ($A=LU$ invertible)

13. Decompositions

- a) LU decomp. for $n \times n$ matrices: $A = LU = LDU$ (Gauss-Elin) (G-J Elin)
- $\rightarrow$ stable way of solving $\hat{x} = A^{-1}b$: $(A\hat{x} = b \rightarrow L\underbrace{Ux}_y = b \rightarrow \underbrace{Ly = b}_{Ux = y})$ } multiplication

Special case: when A is real symmetric $A = LDU = LDL^T = L_1 L_1^T$ (Cholesky decomp.)

$u = L^T \quad L_1 = L\sqrt{D}$

b) QR decomposition ($A_{m \times n}$, full rank)

$$A_{m \times n} = Q_{m \times n} \cdot R_{n \times n}$$

$\rightarrow$ upper triangular matrix with positive diagonal elements

$\rightarrow r_{ij} = \langle q_i, a_j \rangle$ for $i \neq j, i < j$

$[a_1, \dots, a_n]$ $[q_1, \dots, q_n]$

Orthogonal matrix: $Q^T Q = I$

$$\begin{aligned} \rightarrow \hat{x}_{LS} &= (A^T A)^{-1} A^T b \\ &= (R^T Q^T Q R)^{-1} R^T Q^T b = (R^T R)^{-1} R^T Q^T b \end{aligned}$$

c) Eigen decomposition (spectral theorem)

$$A_{n \times n} = Q \cdot D \cdot Q^T \quad (\text{Special case of } A = XDX^{-1})$$

D : contains eigenvalues

Q : contains eigenvectors (Orthogonal matrix)

d) Singular value decomposition (Not to be tested)

$$A_{m \times n} = U_{m \times n} \Sigma_{m \times n} V_{n \times n}$$

$\rightarrow$ close to diagonal: $\Sigma = \begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_n \end{bmatrix}$

U, V orthogonal matrix

$$\sigma_i = \sqrt{\lambda_i} \quad ; i=1, 2, \dots, n$$

λ_i = eigenvalue for $A^T A$